

Application Development Portfolio

1. [A Learning Management System](#)
*Flutter application with HTML/Javascript/Node.js Web Portal.
Masters Capstone Project.*
2. [Overdubber](#)
Native iOS/Swift audio recording application
3. [Japanesify](#)
Mozilla Firefox addon, Javascript
4. [Home Insurance Application Prototype](#)
Native Android Usability Testing Prototype

Find also my bachelor's thesis about digital sound compression here:
<http://publications.lib.chalmers.se/records/fulltext/251862/251862.pdf>

A Learning Management System.

Capstone Project. Project title: App for Android Students.
Masters of IT (Professional Computing), Software Development
Swinburne University of Technology

Completion 2020-06-02.

Development team: Bilal Hassan, Sabin Maharjan and Thorbjørn Bonvik

This application is a learning management system intended for an Android course taught at Swinburne University.

Video Presentation

<https://drive.google.com/file/d/1Glt4dJ9230Xb4oM2Nh2ZMgIPf22EgdRN/view?usp=sharing>

List of features:

Mobile app:

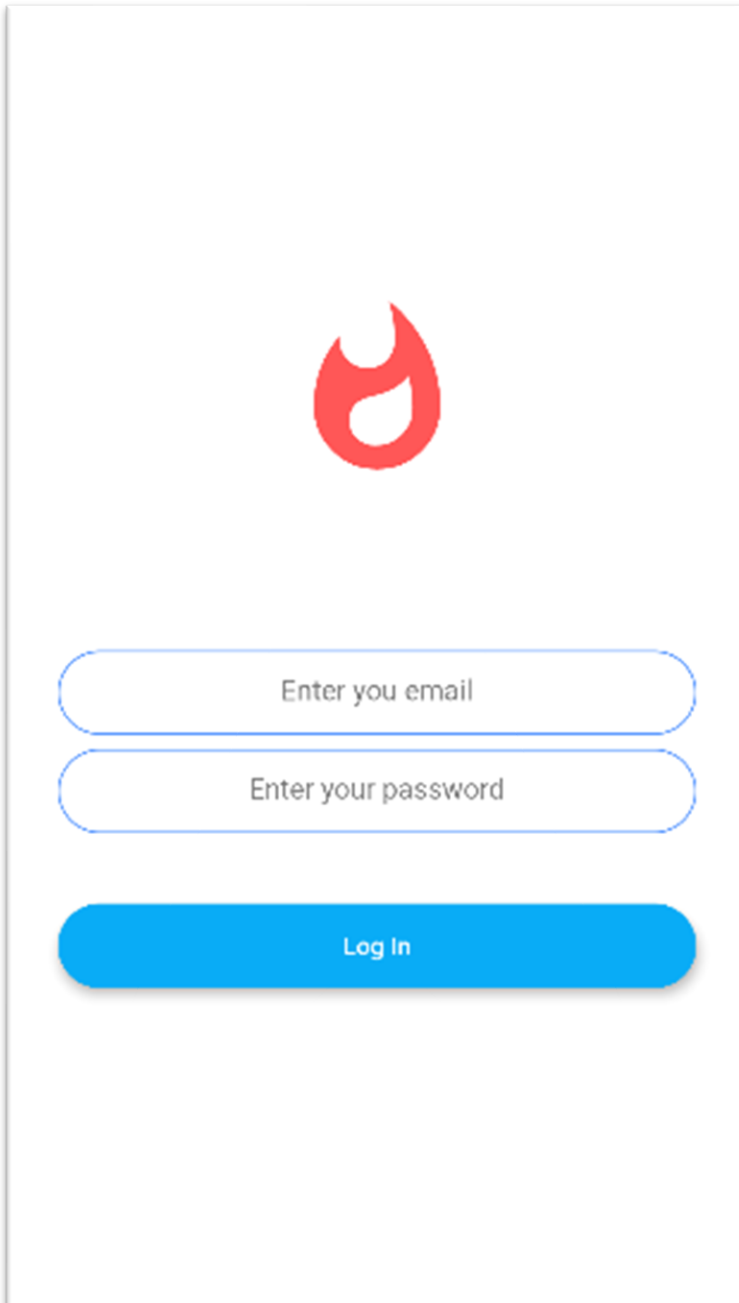
- Login Page
- Home Screen Display
 - Containing User Profile, Today's Lectures and Tasks that are due soon.
- Schedules Tab
 - Takes to schedule details on clicking item.
 - User can send star rating and review for the lecture/workshop content.
- Tasks Tab
 - Takes to Task details on clicking item
 - User can change the task marked completed or incomplete
- Account Tab
 - User can view the profile details.
 - User can change the target grade.
- Notifications
 - App shows notifications for schedule and tasks due dates
 - App receives and shows notification on schedule or task update from web portal.

Webportal

1. List, Create, Update and Delete Schedules
2. List, Create, Update and Delete Tasks
3. List, Create, Update and Delete Students
4. Send notifications

Screenshots

Login

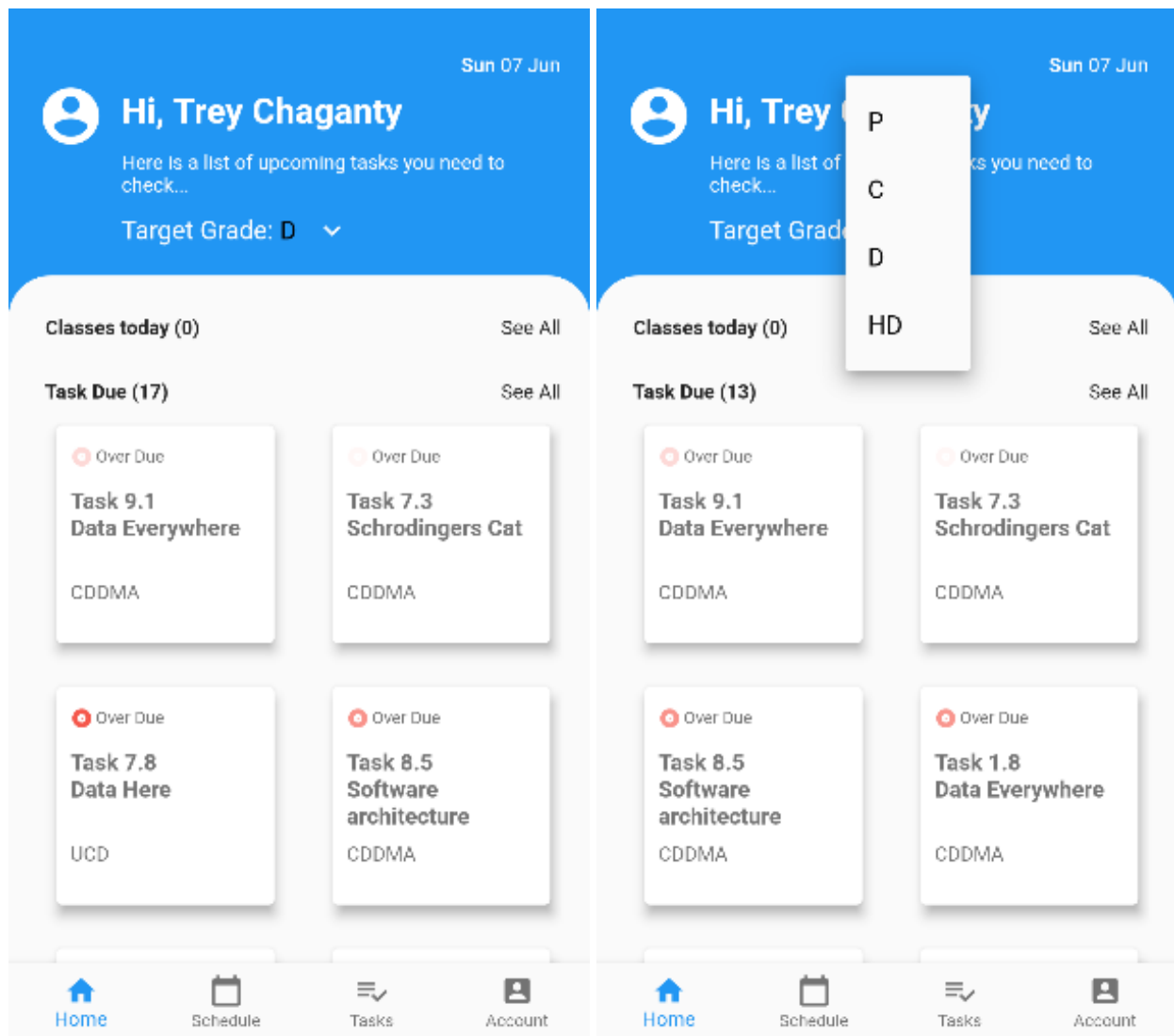


The screenshot shows a login interface. At the top center is a red flame logo. Below it are two input fields: "Enter you email" and "Enter your password". At the bottom is a blue "Log In" button.

The first screen that shows up on the installation.

Use the login credentials provided by the web portal.

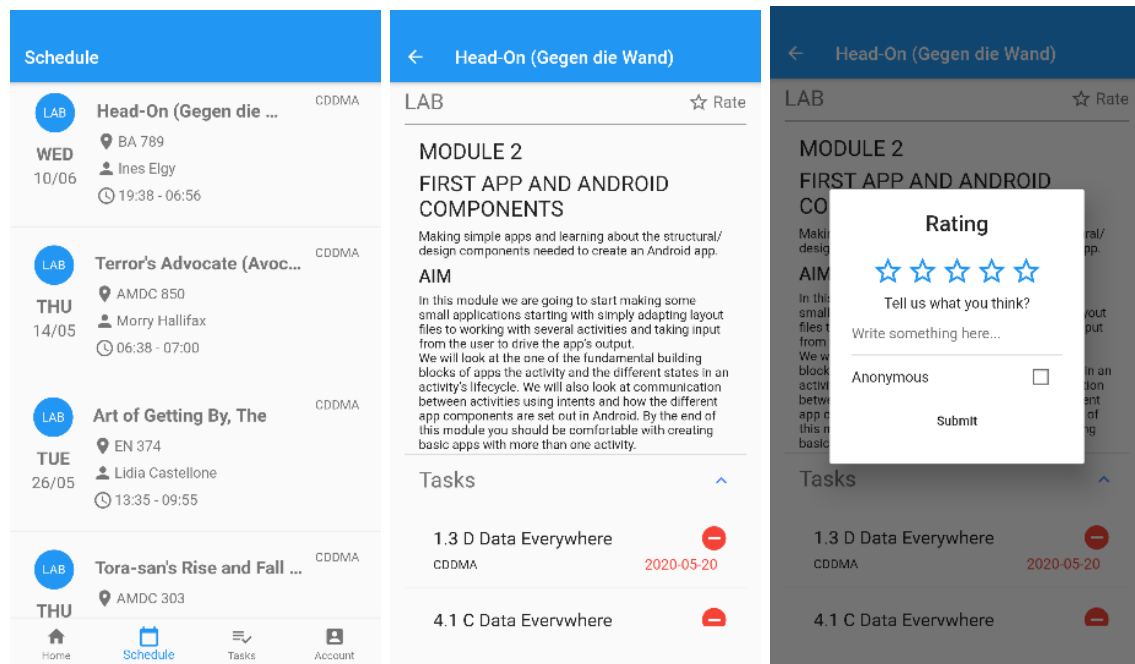
Home View



Home Screen after a successful login.

- Click the target Grade and change to desired one.
- Click listed classes to view the class details.
- Click see all button in front of Classes Today to open all classes list tab (Schedule tab).
- Click any Task grid item and this will take you to its details page.

Schedule Tab & Schedule Details & Rating option



The schedules page displays all the schedules for the selected target grade by the student.

-Scroll to view all the schedules.

-Click on a schedule and this will take you to details page.

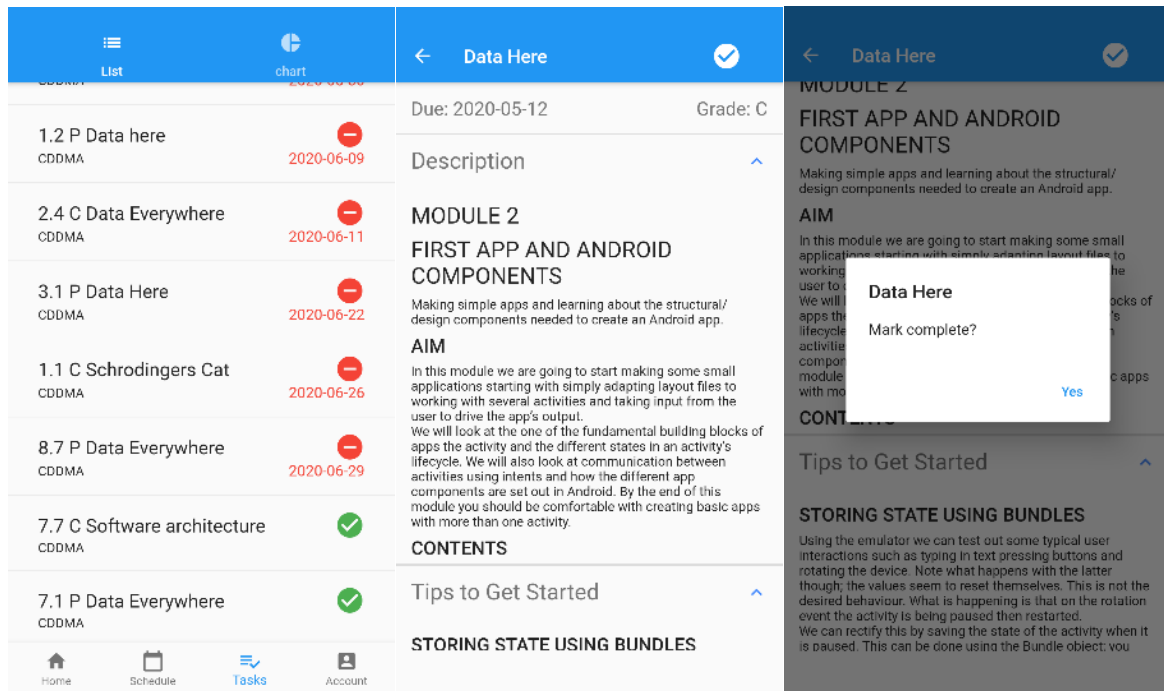
-Scroll to view description of the class or lab.

-Tap on the Tasks heading to view linked tasks to the lecture or lab.

-Click any task to view details of the task.

-Click the rate option at top right corner to send ratings and review for the lecture or lab.

Tasks Tab & Task details & Mark-offs



Tasks tab displays all the tasks that user has marked complete or not.

-Scroll through the list to view completed task at the bottom of the list.

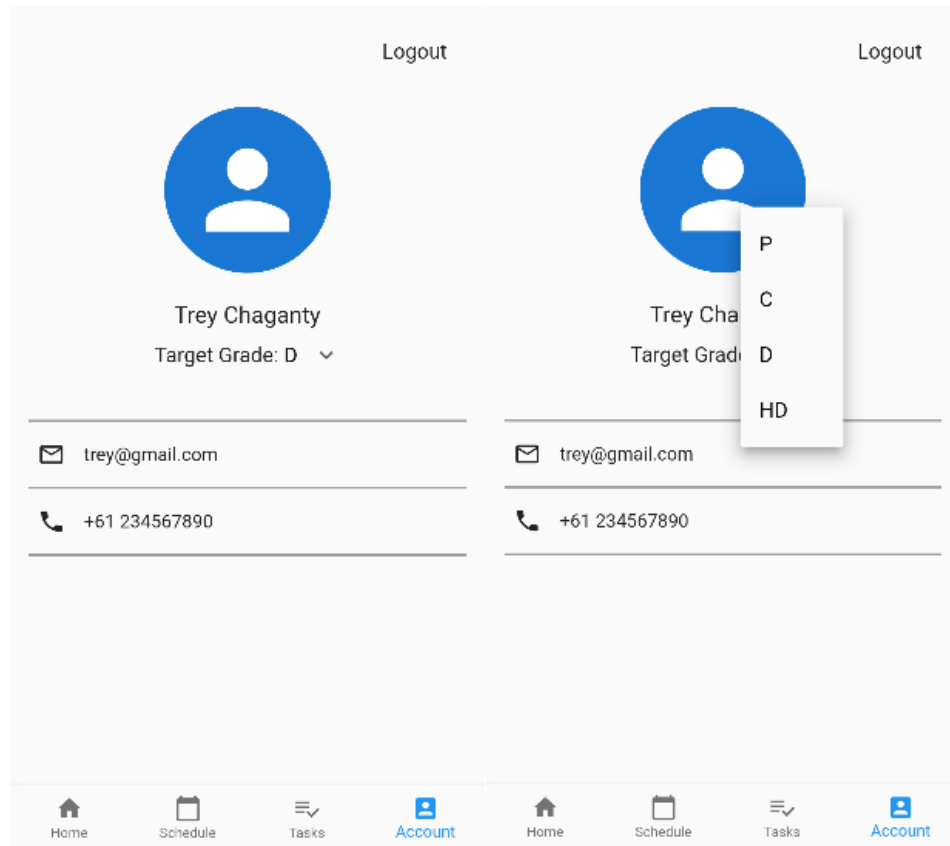
-Click the item to view the details of that task.

-Scroll to view description of the task.

-Tap on the tips heading to expand the tips details and scroll to read through.

-Click the button on the top right corner to set the task complete or incomplete.

Account View

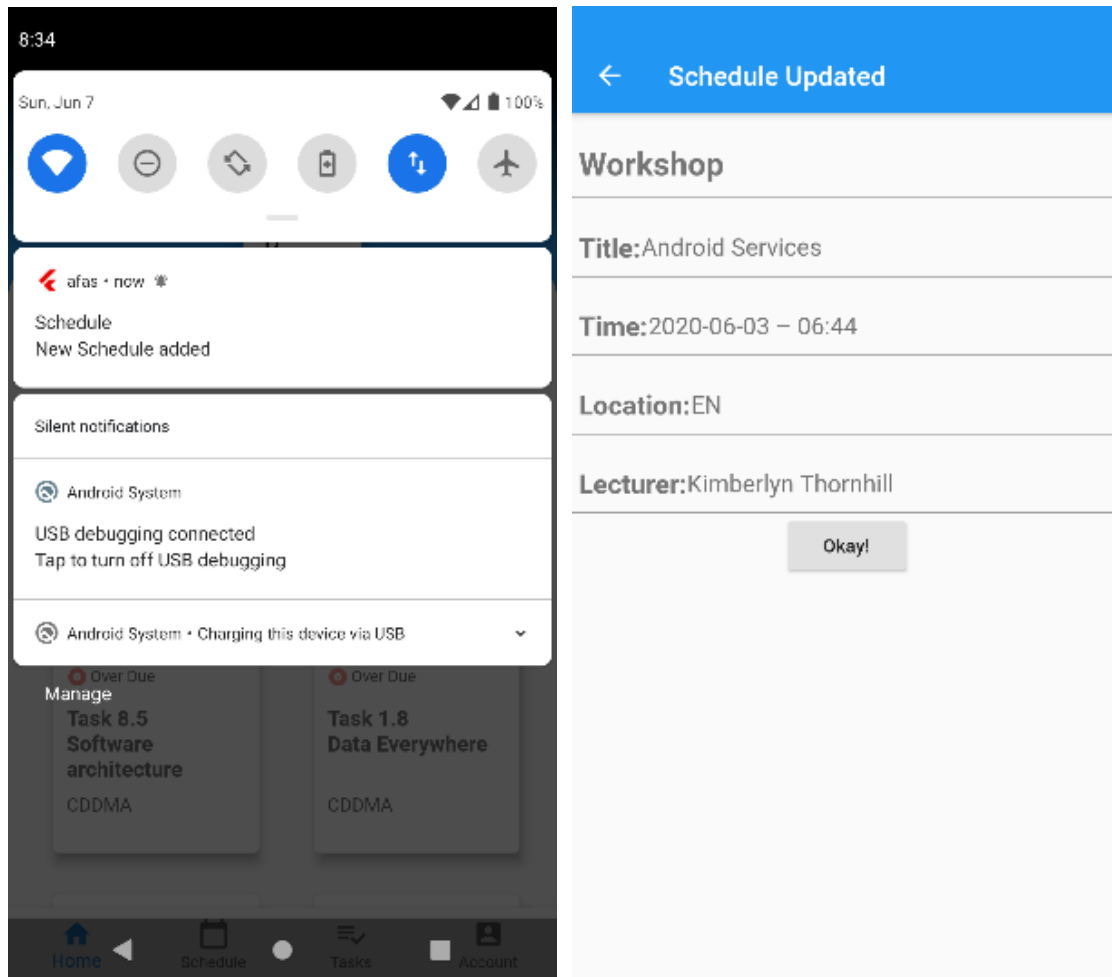


Account View tab display user registered details and logout button.

-User can only change the target grade and all other details are in control of the teacher's portal.

-Click logout button at top right corner to logout and get back to login screen.

Notifications



Notifications are displayed and received when a schedule or task update is made on the teacher's portal or from the schedules that app already has.

- For schedule a notification will be displayed at the starting time of the lecture or lab.

- For tasks, notifications will be displayed a day and an hour before the due time.

- On clicking the notification, the details of notification will be displayed.

- On schedule or task update from teacher portal, the list at mobile app will be updated automatically.

Schedules

List Schedules:

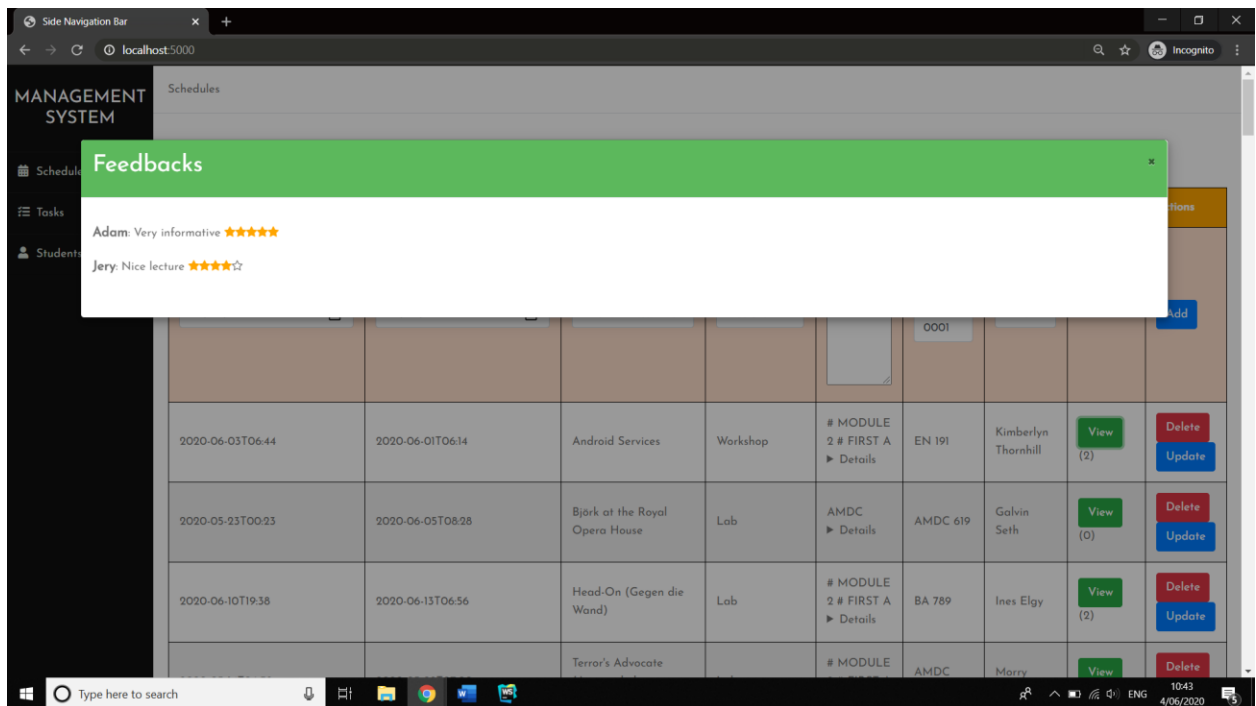
- The list of schedules can be viewed by clicking schedules menu button from the left side of the screen.

The screenshot shows a web application titled 'MANAGEMENT SYSTEM' with a sidebar menu containing 'Schedules', 'Tasks', and 'Students'. The main content area is titled 'Manage Schedule' and displays a table of schedules. The table has the following columns: Time Start, Time End, Title, Type, Notes (md), Room, Tutor, Feedbacks, and Actions.

Time Start	Time End	Title	Type	Notes (md)	Room	Tutor	Feedbacks	Actions
02/06/2020 10:30	02/06/2020 12:30	This Is A New Even	Lecture		AMDC 0001	Dr. Smit		Add
2020-06-03T06:44	2020-06-01T06:14	Android Services	Workshop	# MODULE 2 # FIRST A ► Details	EN 191	Kimberlyn Thornhill	View (2)	Delete Update
2020-05-23T00:23	2020-06-05T08:28	Björk at the Royal Opera House	Lab	AMDC ► Details	AMDC 619	Galvin Seth	View (0)	Delete Update
2020-06-10T19:38	2020-06-13T06:56	Head-On (Gegen die Wand)	Lab	# MODULE 2 # FIRST A ► Details	BA 789	Ines Elgy	View (2)	Delete Update
		Terror's Advocate		# MODULE	AMDC	Marry	View	Delete

View feedbacks:

- The feedbacks for each schedule can be viewed by clicking the [view](#) button on the respective schedule.



Create a Schedule:

- You can create a schedule by typing in
 - start time
 - end time
 - schedule title
 - Schedule Type: Can be workshop, tutorial, lab or lecture
 - Room: Building with room number
 - Notes: The text in markdown format
 - Tutor
- And click the **Add** button

Manage Schedule

Time Start	Time End	Title	Type	Notes (md)	Room	Tutor	Feedbacks	Actions
02/06/2020 10:30	02/06/2020 12:30	Android Componen	Lecture	Apps consist of Activities which contain Views. Views are di	AMDC 101	Dr. Smit		Add
2020-06-10T19:43:51Z	2020-06-21T17:00:38Z	(Prime Time) (Funny America)	tutorial	2 # FIRST A ► Details	EN 440	Hirsch Gratex	View (0)	Delete Update
2020-06-02T10:30	2020-06-02T12:30	Android Componenets	Lecture	# MODULE 2 # FIRST A ► Details	AMDC 101	Dr. Smith	View (0)	Delete Update

Note: The new schedule will be added at the bottom of the list.

Update a Schedule:

1. You can edit/update a current schedule by clicking on **Update** button on the schedule which you wish to edit.
2. After you click the **Update** button, the schedule that you wish to edit will show at the top of the table in the editable text fields.
3. You can change the values that you want to change there.
4. If you are happy with the change, click **Commit Update** button. The schedule will be updated.
5. If you do not wish to change, click **Cancel Update** button.

Manage Schedule

Time Start	Time End	Title	Type	Notes (md)	Room	Tutor	Feedbacks	Actions
02/06/2020 10:30	02/06/2020 12:30	Android Componen	Lecture	# MODULE 2 # FIRST APP AND ANDROID	AMDC 101	Dr. Smit		Commit Update Cancel Update
				# MODULE			Yes	Delete

Delete a schedule:

1. You can delete a schedule by click a **Delete** button on the schedule you wish to delete.
2. An alert box will pop-up to confirm the delete action.
3. Select **OK** to delete the schedule.
4. Select **Cancel** if you do not want to delete the schedule.

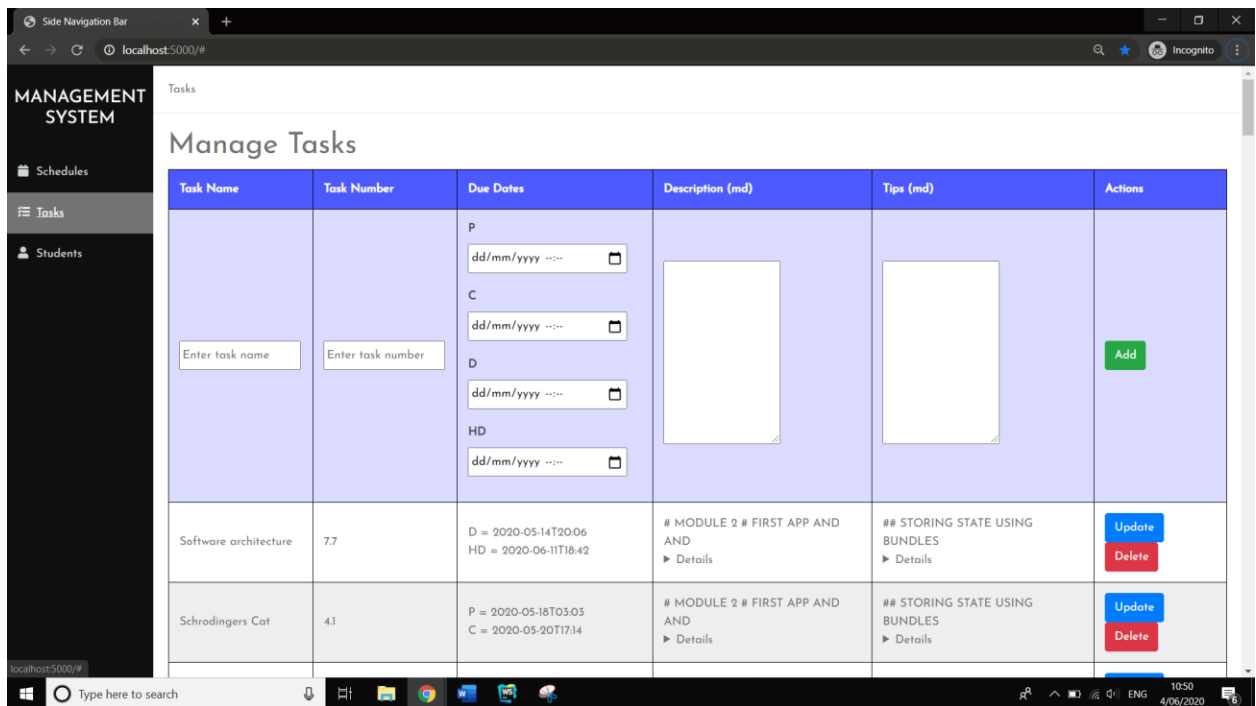


Note: If you cannot see the new item, or modified item, please refresh the page.

Tasks

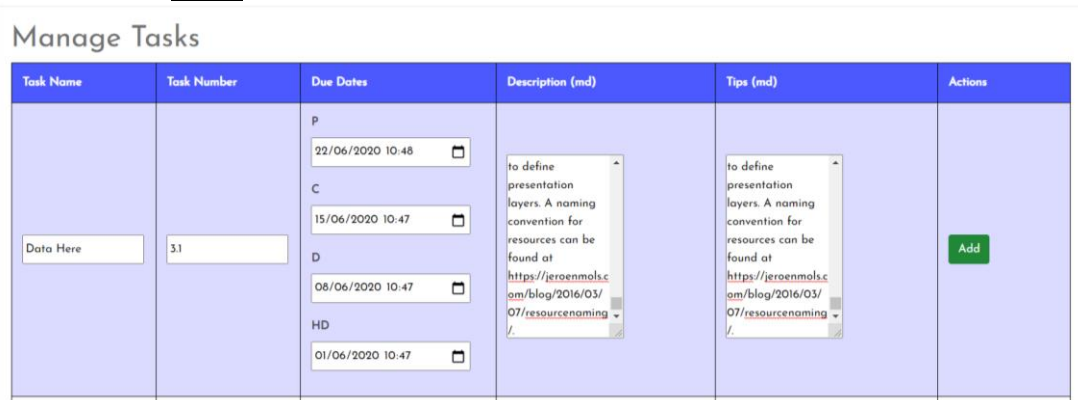
List Tasks:

- The list of tasks can be viewed by clicking tasks menu button from the left side of the screen.



Create a Task:

- You can create a task by typing in
 - Task name
 - Task number
 - Due dates: The different due dates can be set for different task grade.
 - Description: The text in markdown format
 - Tips: The text in markdown format
- And click the **Add** button



Note: The new task will be added at the bottom of the list.

		C = 2020-06-02T11:16	► Details	► Details	Delete
Data Here	3.1	P = 2020-06-22T10:48 C = 2020-06-15T10:47 D = 2020-06-08T10:47 HD = 2020-06-01T10:47	# MODULE 2 # FIRST APP AND AND ► Details	# MODULE 2 # FIRST APP AND AND ► Details	Update Delete

Update a Task:

1. You can edit/update a current task by clicking on **Update** button on the task which you wish to edit.
2. After you click the **Update** button, the task that you wish to edit will show at the top of the table in the editable text fields.
3. You can change the values that you want to change there.
4. If you are happy with the change, click **Update** button. The task will be updated.
5. If you do not wish to change, instead add a new task from the current data, click **Add** button.

Manage Tasks

Task Name	Task Number	Due Dates	Description (md)	Tips (md)	Actions
Data Here	3.1	P 22/06/2020 10:48 C 15/06/2020 10:47 D 08/06/2020 10:47 HD 01/06/2020 10:47	# MODULE 2 # FIRST APP AND ANDROID COMPONENTS Making simple apps and learning about the structural/design components needed to create	# MODULE 2 # FIRST APP AND ANDROID COMPONENTS Making simple apps and learning about the structural/design components needed to create	Add Update

Delete a task:

1. You can delete a task by click a **Delete** button on the task you wish to delete.
2. An alert box will pop-up to confirm the delete action.
3. Select **OK** to delete the task.
4. Select **Cancel** if you do not want to delete the task.

localhost:5000 says
Are you sure to delete this record ?

OKCancel

Note: If you cannot see the new item, or modified item, please refresh the page.

Students

List Students:

- The list of students can be viewed by clicking students menu button from the left side of the screen.

Students

Manage Students

Name	Email ID	Password	Target Grade	Actions
Enter name	Enter email	Enter password	P	Add
Almeda Pues	apues0@globa.com	7gluBafIUPlk	D	Update Delete
Kathryne Runnalls	Kathryne@globa.com	7gluBafIUPlk	D	Update Delete
Trey Choganty	trey@gmail.com	abi12	D	Update Delete
Cordie Schwacha	Cordie@gmail.com	7gluBafIUPlk	HD	Update Delete
Bilal Hassan	bilalhx@gmail.com	abc123	D	Update Delete
Rami Malek	12345678@student.swin.edu.au	12345678	C	Update Delete

Create a Student:

- You can create a student by typing in
 - Student name
 - Student email id
 - Student password
 - Target Grade: The grade that students want to achieve. This is by default P, as students should at least have a P grade to pass the unit. Students can change this via the mobile app.
 - Due dates: The different due dates can be set for different task grade.
- And click the **Add** button

Manage Students

Name	Email ID	Password	Target Grade	Actions
new student	12341234@student.swin	12341234	P	Add

Note: The new student will be added at the bottom of the list.

new student	12341234@student.swin.edu.au	12341234	P	Update Delete
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Update a Student:

1. You can edit/update a current student by clicking on **Update** button on the student which you wish to edit.
2. After you click the **Update** button, the student that you wish to edit will show at the top of the table in the editable text fields.
3. You can change the values that you want to change there.
4. If you are happy with the change, click **Update** button. The student will be updated.
5. If you do not wish to change, instead add a new student from the current data, click **Add** button.

Manage Students

Name	Email ID	Password	Target Grade	Actions
new student	12341234@student.swin	12341234	P	Add Update

Delete a student:

1. You can delete a student by click a **Delete** button on the student you wish to delete.
2. An alert box will pop-up to confirm the delete action.
3. Select **OK** to delete the student.
4. Select **Cancel** if you do not want to delete the student.

localhost:5000 says

Are you sure to delete this record ?

Note: If you cannot see the new item, or modified item, please refresh the page.

Notifications

The notification will automatically be sent from the website when the user adds a schedule or a task.

Overdubber

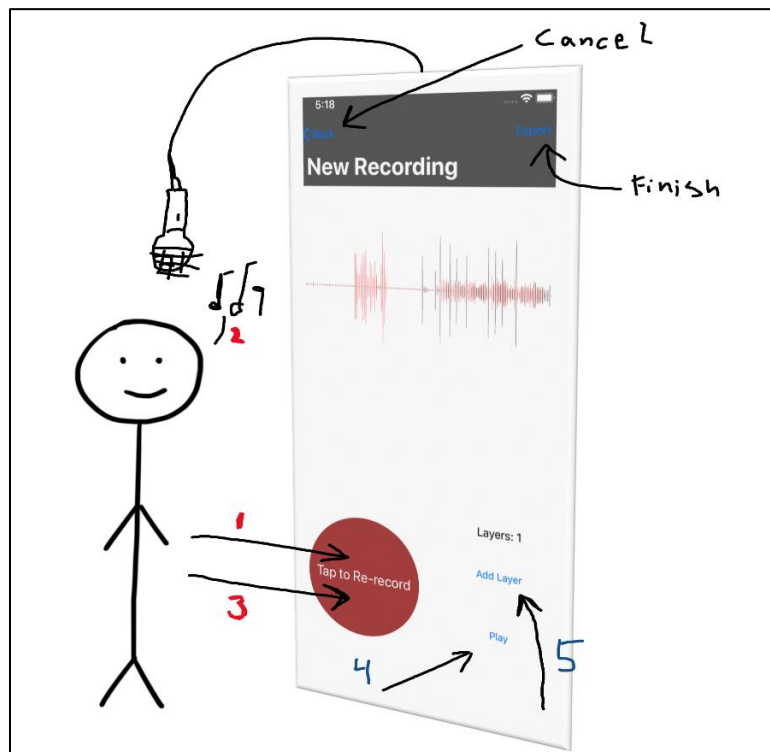
Data Driven Mobile Applications course final project / Passion project

Development team: Thorbjørn Bonvik

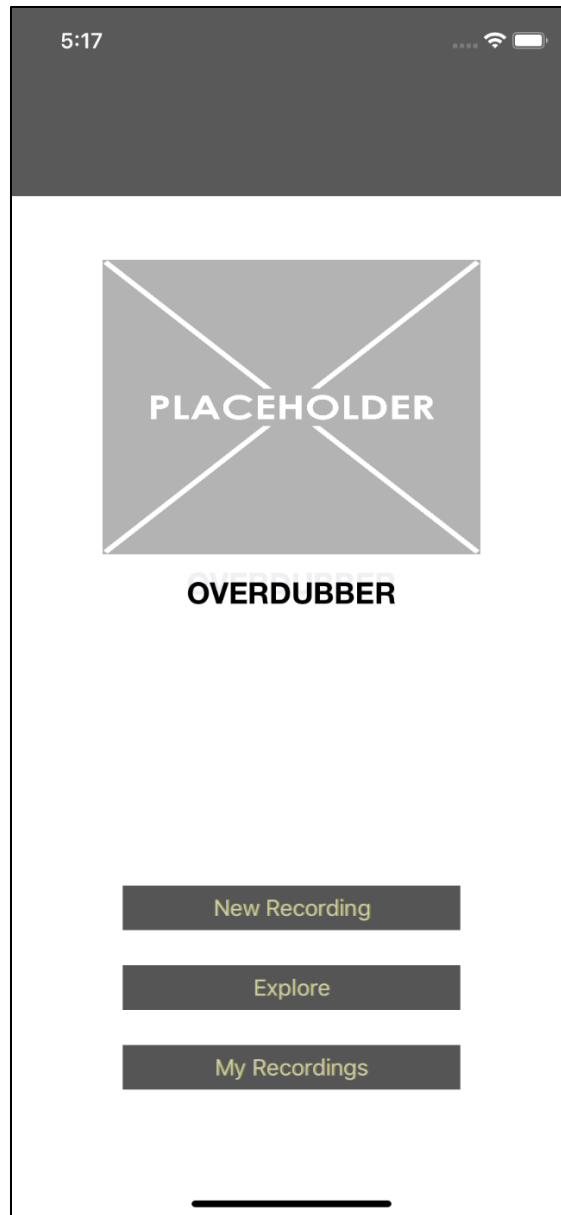
This Swift/iOS application is a recording application that can record multiple takes and merge them into a single file. The intended users are musicians that want to create a lofi or quick recording of a track and children that want to explore the joys of recording and creating.

Note that this was created for a project focusing on sensors and data modification in mobile development, and the UI is not the prettiest.

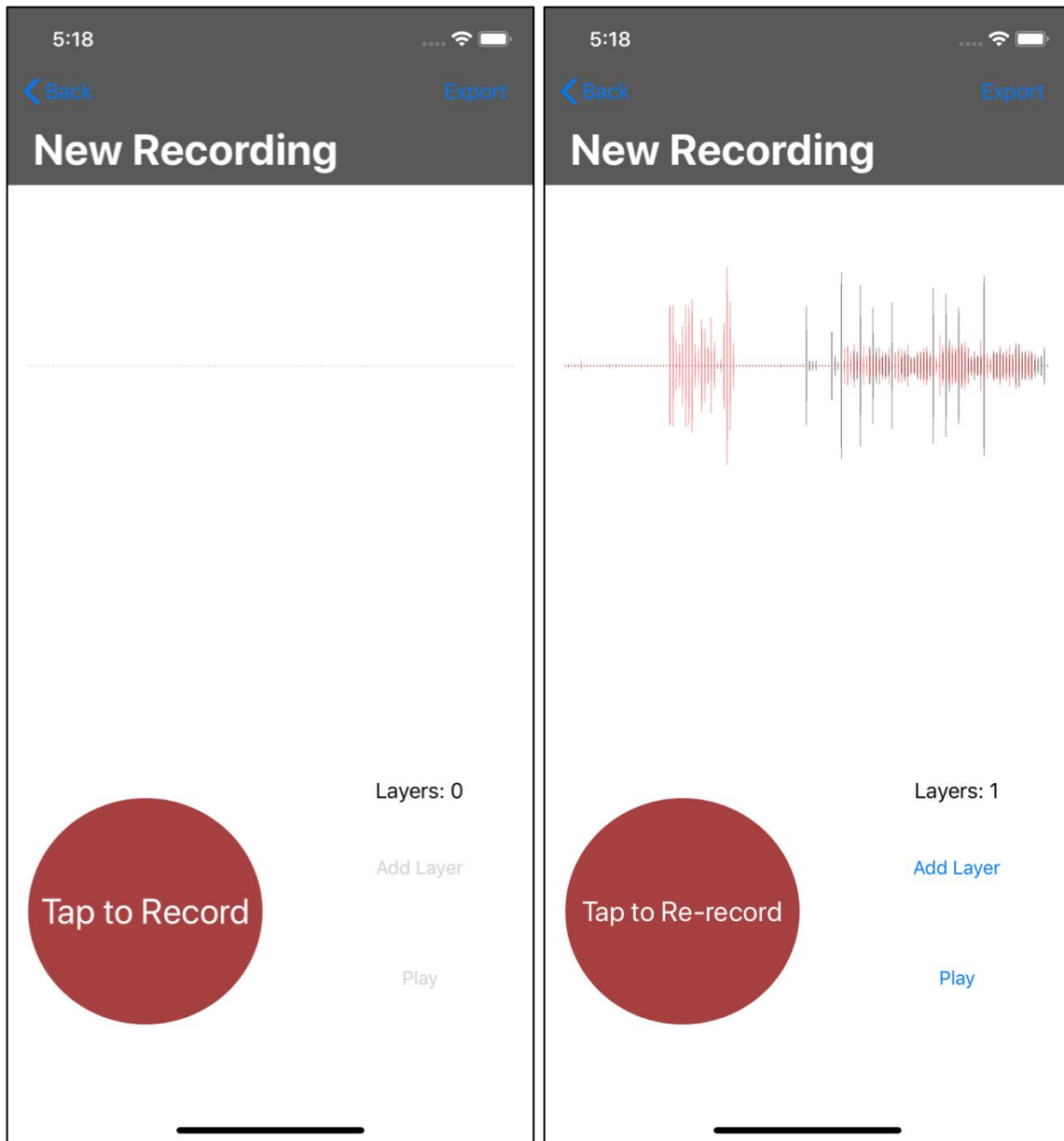
After an UI update and Soundcloud integration, this will be released to the App store.



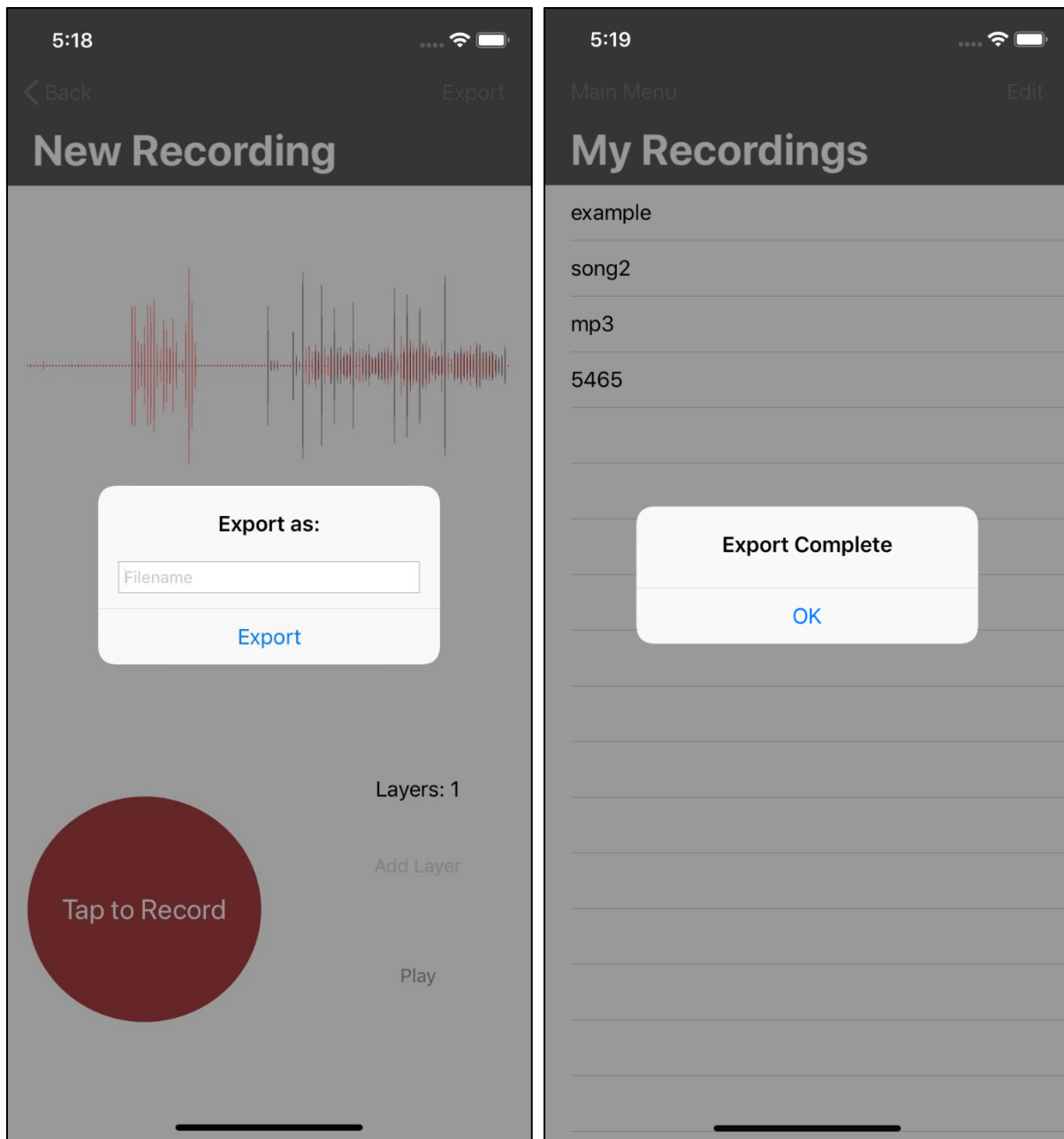
User Interaction



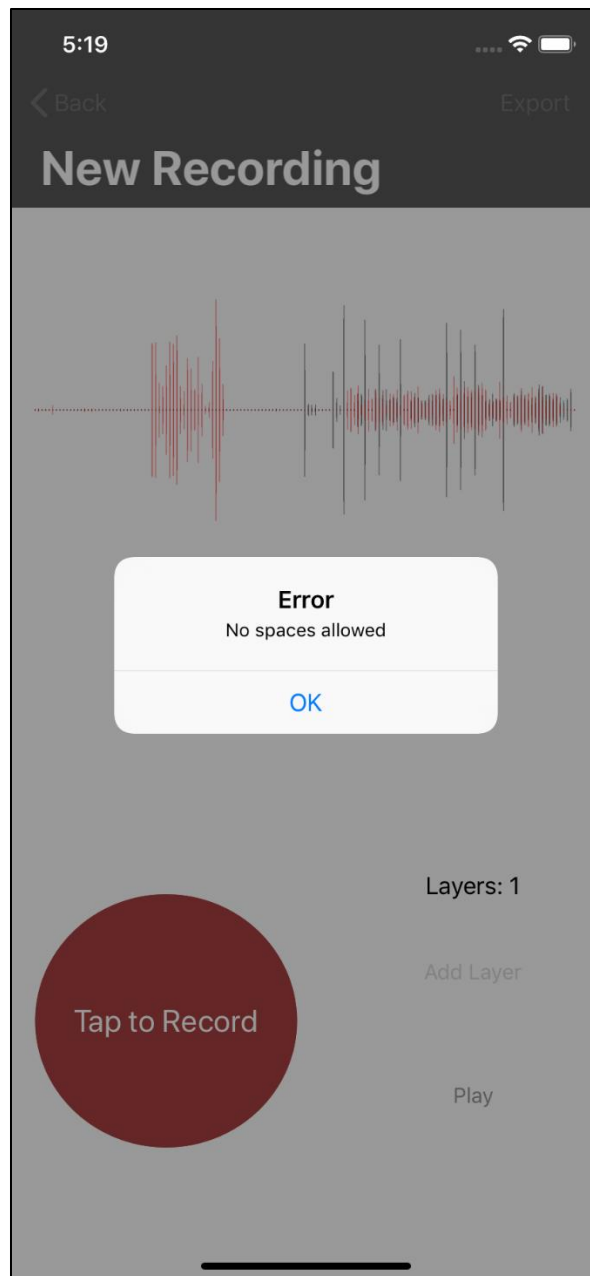
Main Menu



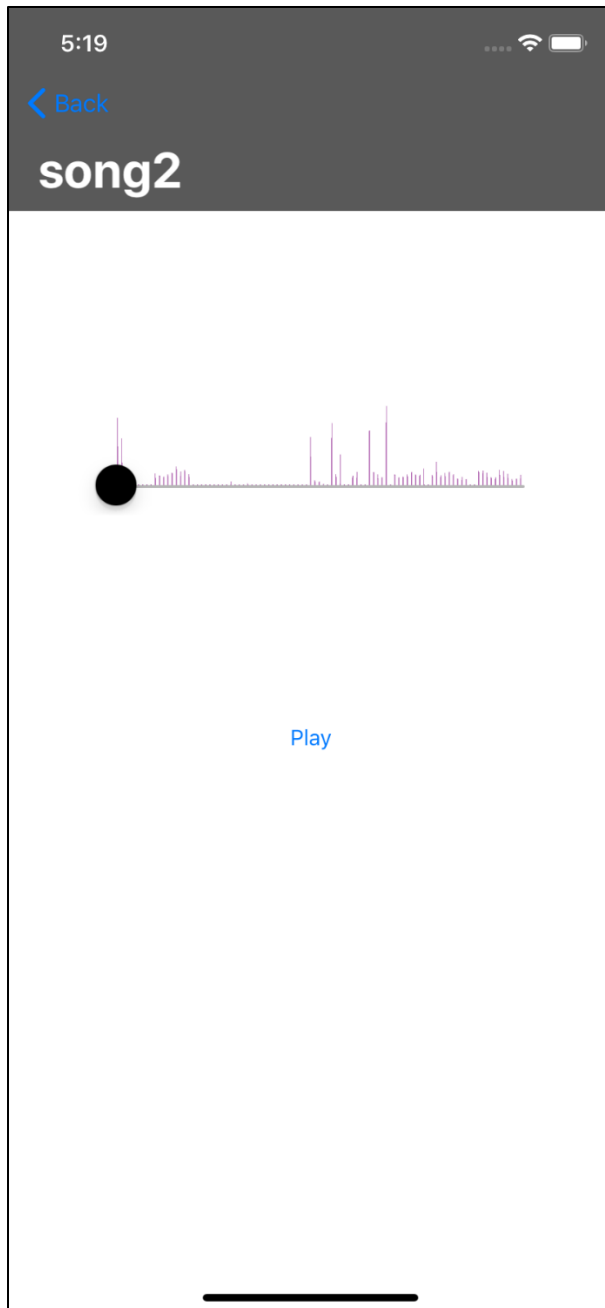
Recording: No takes vs 2 takes



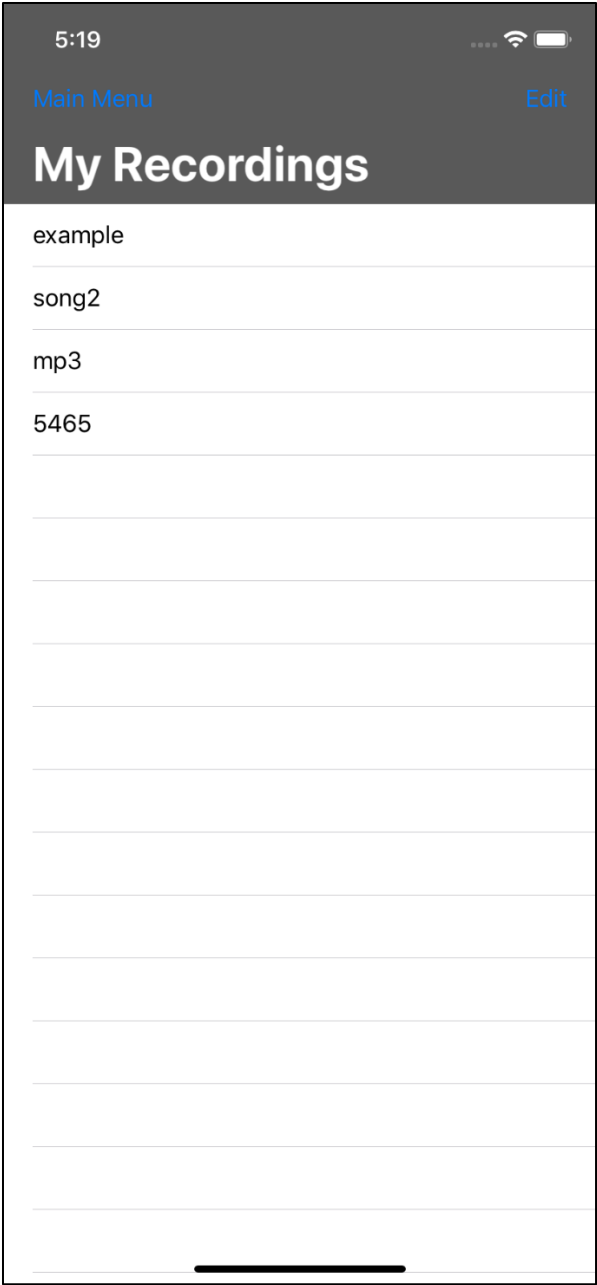
Successful export



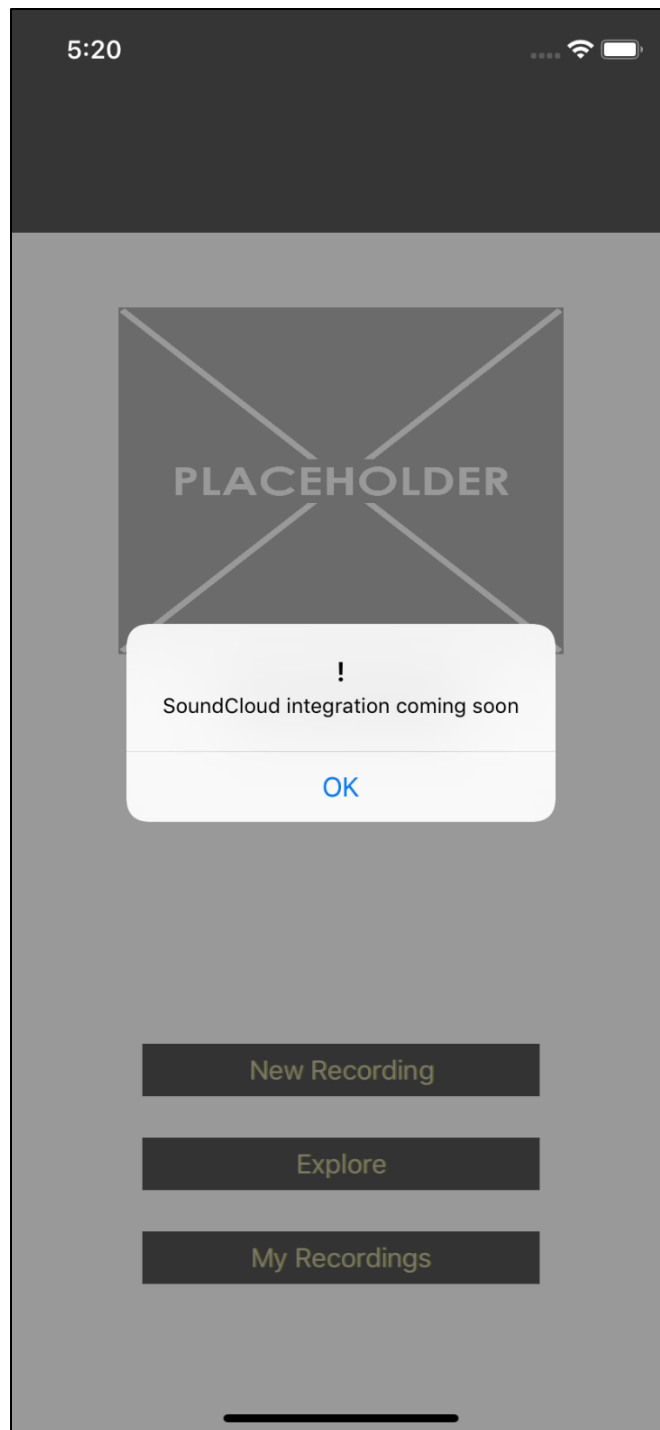
File export error



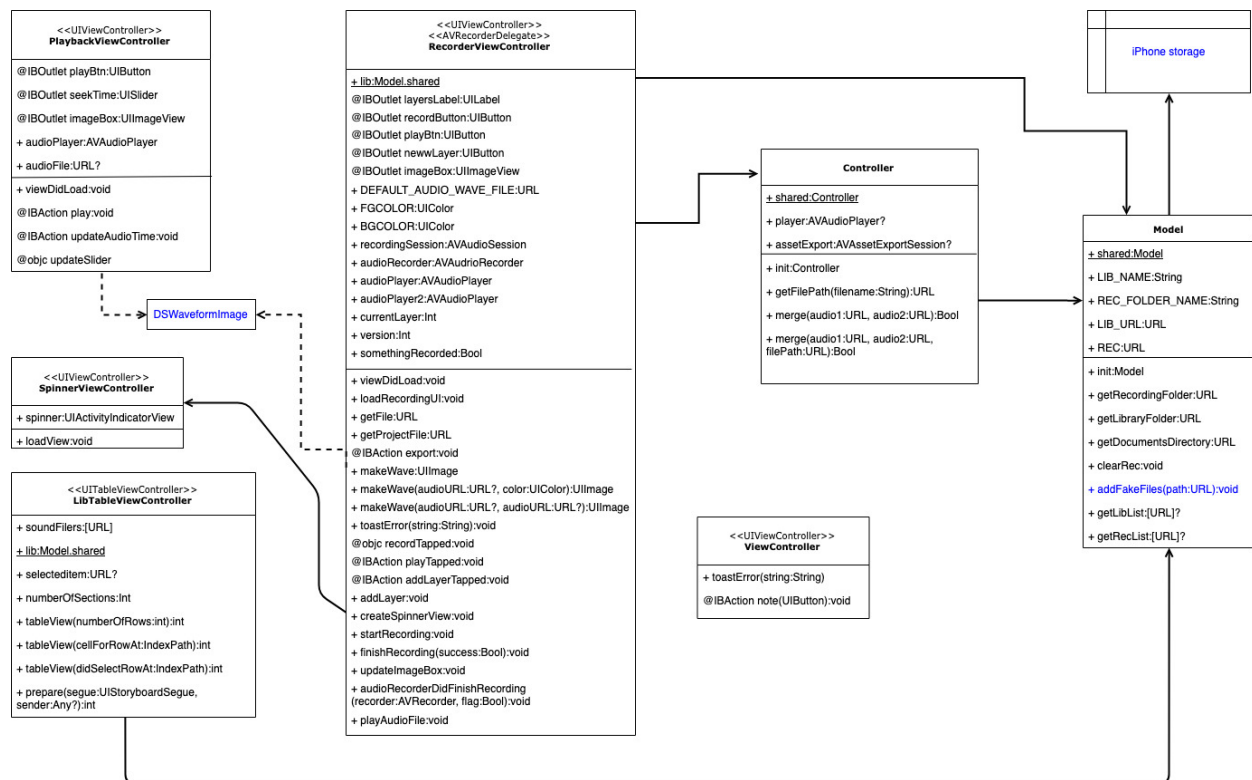
Playback screen



My Recordings



“Explore”



Class Diagram

Repository: <https://github.com/thorbjxrn/overdubber>

Japanesify webextension.

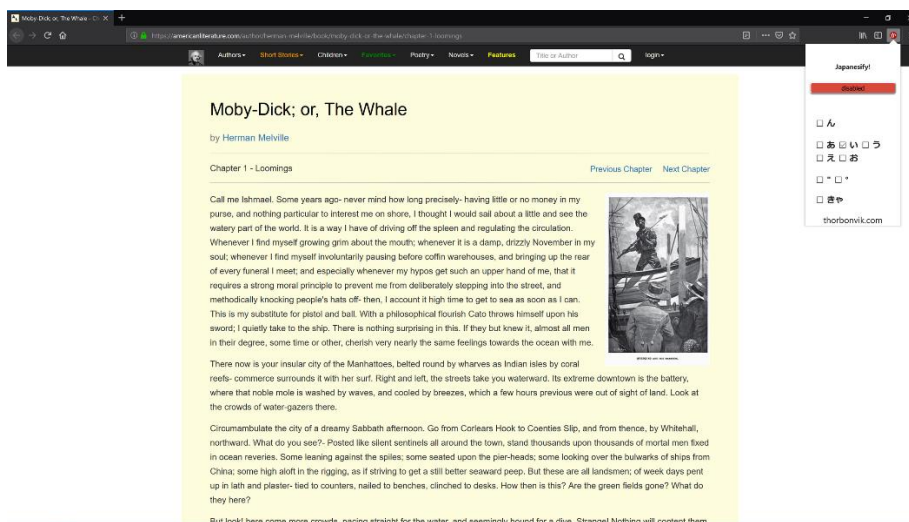
Open Source Passion project

Development team: Thorbjørn Bonvik and "hcortizcolon".

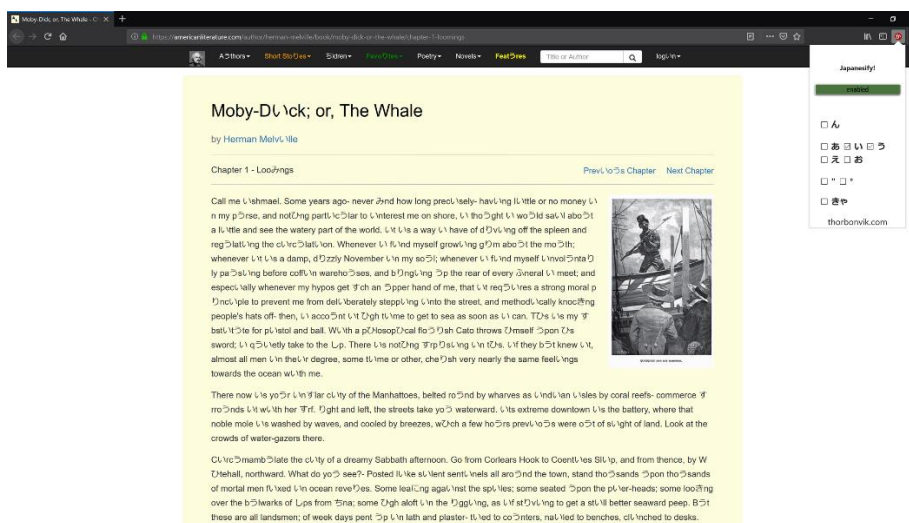
A WebExtension that will replace latin letters with the selected Japanese characters. The idea is to gradually expand and practice the Japanese alphabets without intense study sessions. This WebExtension uses local storage to save preferences.

This can be installed from Mozillas repositories.

<https://addons.mozilla.org/en-US/firefox/addon/japanesify/>



Disabled

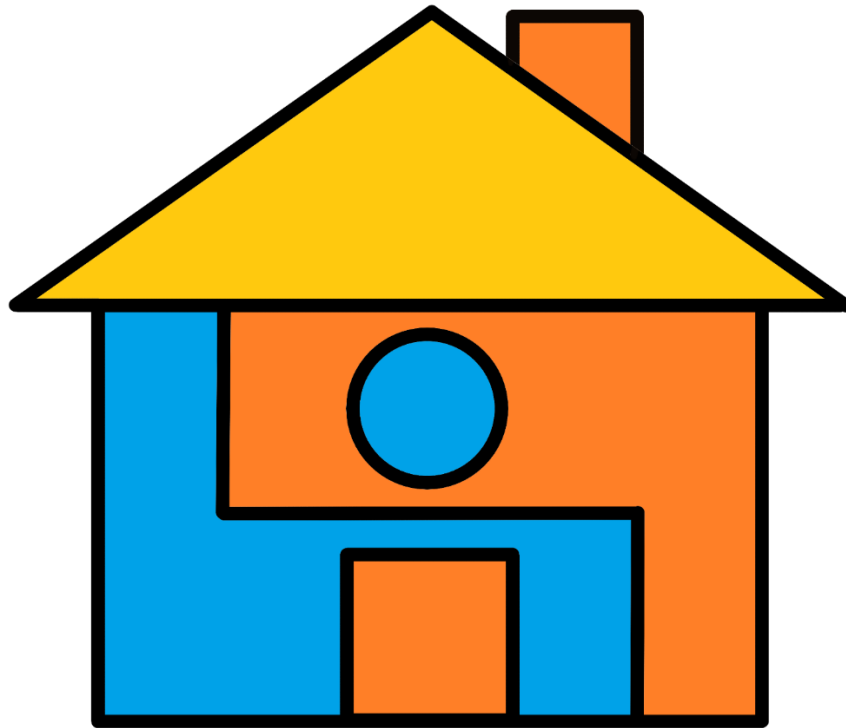


Enabled

Repository: <https://github.com/thorbjorn/overdubber>

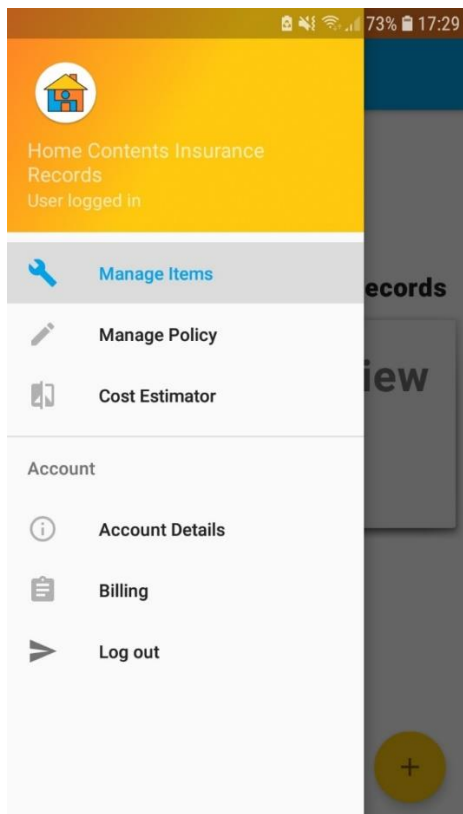
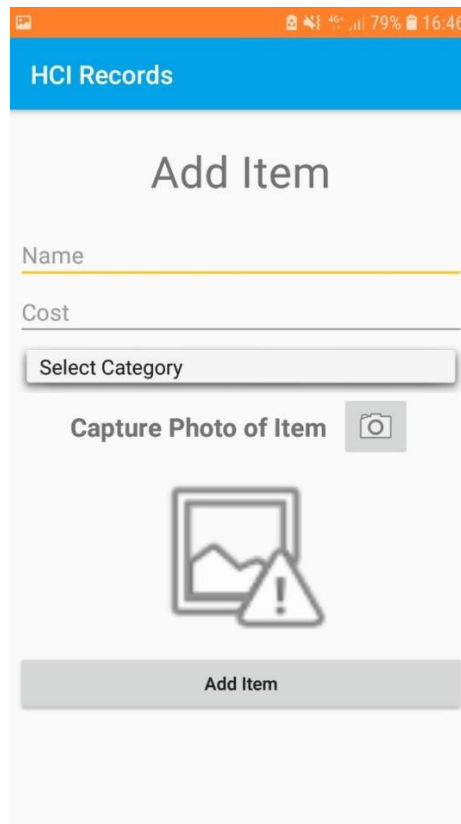
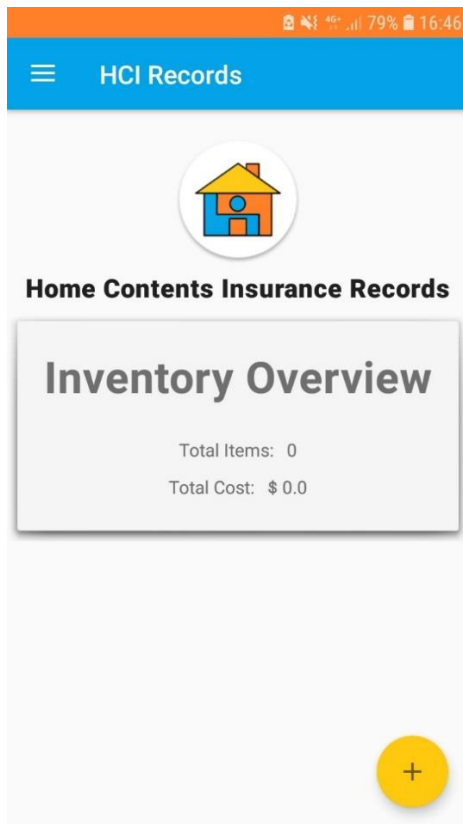
Home Contents Insurance

This application was created as a User Centered Design prototype. It is the result of and subject of continuous iterative development cycles with user centered evaluation.



Icon designed by Thorbjørn Bonvik for a home insurance application.
Chaos and order, fire and water, man and technology.

Development team: Bilal Hassan and Thorbjørn Bonvik



Screenshots